The Petrochemistry and Paleomagnetism of the Molson Dikes, Manitoba

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Northeasterly trending, tholeiitic, low-alkali basaltic dikes which cut the Archean rocks of northeastern Manitoba have yielded a paleomagnetic pole at 114°W, 28°N, α_{95} = 12°. The material from nine sites was normally magnetized, and from one site, reversely magnetized. Potassium–argon age determinations suggest that the dikes were magnetized during late Aphebian. They may have been magnetized at the time of intrusion. Strongly differentiated dikes show intense and variable deuteric alteration, which is interpreted to be the cause of within-site dispersion of NRM. However, strong NRM is related to high contents of opaques regardless of the degree of alteration. Aphanitic dikes of low magnetite content and exhibiting minimal magmatic differentiation usually record moderate magnetic intensities and good groupings.

Des dykes tholéitiques et basaltiques à faible teneur en alcali d'attitude nord-est coupant les roches archéennes du nord-est manitobain produisent un pôle magnétique à 114°W, 28°N, α_{95} = 12°. Les échantillons provenant de 9 sites étaient normalement magnétisés alors que celui d'un autre site montrait un magnétisme inversé. Les déterminations d'âge par la méthode potassium-argon suggèrent que les dykes furent magnétisés pendant l'Aphébien supérieur. Il est probable qu'ils aient été magnétisés lors de l'intrusion. Des dykes fortement différenciés montrent des altérations deutériques intenses et variables que l'on croit être la cause des dispersions internes du NRM. Cependant, un fort NRM est relié au contenu élevé en minéraux opaques indifféremment du degré d'altération. Des dykes aphanitiques à faible contenu en magnétite et montrant des différences magnétiques minimes enregistrent ordinairement des intensités modérées et présentent de bons groupements. [Traduit par le journal]

Introduction

The Molson dikes constitute a group of northeasterly trending diabase intrusions within the Superior Orogen of Northern Manitoba (Fig. 1). Magnetic data from a few samples of these dikes have been reported (Fahrig *et al.* 1965), but these samples could not be adequately cleaned with instruments then available, so a more extensive collection has been obtained and this was treated with modern demagnetizing equipment. The pole for the Molson dikes is of particular interest as these dikes lie in the Superior Orogen, an area which has not undergone appreciable tectonic disturbance for 2.5 b.y., and which constitutes the reference plate for North American Precambrian paleomagnetism. A reasonably detailed petrochemical study has been carried out to provide basic chemical data on these mantle-derived plateau basalts of Paleohelikian (or possibly Aphebian) age and to consider possible reasons for variation in the magnetic stability of the samples.

Paleomagnetism

Eighty cores were drilled and oriented in the field with solar compass at 12 of the 13 sites, whose locations are indicated by the numbers on Fig. 1. Only chemical and petrographic data are available for material collected at site 12. Two specimens were cut from each core, and the mean direction of magnetization of the two specimens was used to represent the core, except when θ, the angle between the magnetization directions of the two specimens exceeded 25° after optimum cleaning. In the latter case, the core was considered to be inhomogeneously magnetized and data from it were not used in the site statistics.

The optimum cleaning field for each site was obtained by first step-cleaning two specimens per site. All specimens from each site were then cleaned in five steps; from one step below, to well above the optimum field suggested from results obtained on the two test specimens. The optimum cleaning field was then considered to be the one yielding the best grouping for each site.

The NMR directions were rather dispersed (Table 1), but their mean was well away from the recent earth's field direction, suggesting the presence of a stable magnetic component. The within-site dispersion decreased in most sites as a result of AF cleaning (Table 1). However, the

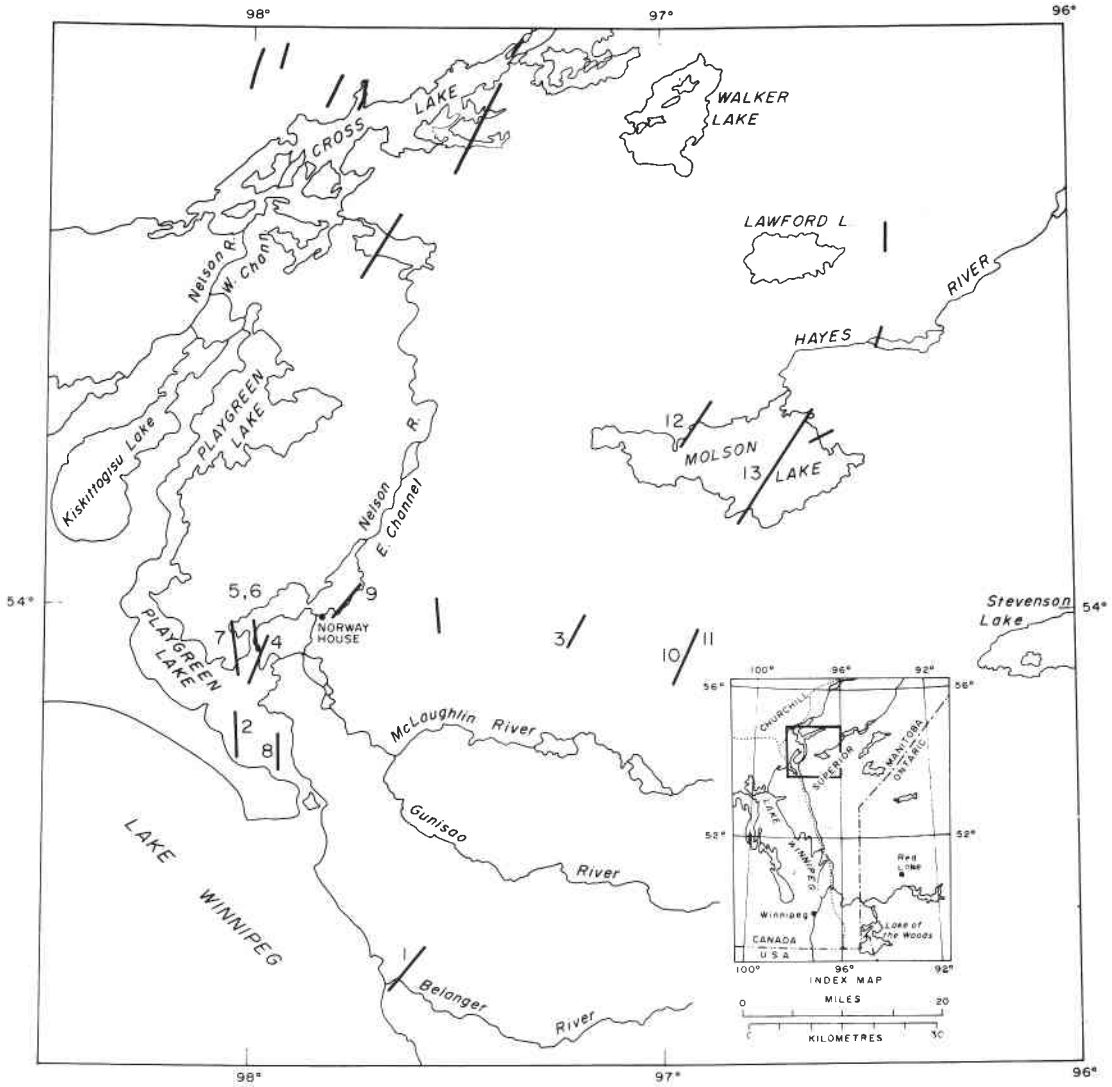


FIG. 1. Distribution of Molson diabase dikes taken from the Cross Lake map-area (Bell 1962) and the Norway House – Grand Rapids map-area (Ermanovics 1973). Size of dikes is grossly exaggerated. Numbers refer to sites discussed in text.

NRM of site 11 was random and remained so during the step-cleaning. The initial intensity of this site and of site 4 was 10^2 to 10^4 times higher than that of the other sites (Fig. 2). Site 4 yielded only three homogeneously magnetized samples, and in the absence of either chemical or petrographic evidence that might explain this, it is suggested that the rock at both sites 11 and 4 were affected by lightning strikes. This is supported by the fact that site 10, which is on the same dike as site 11, exhibited normal initial intensities and every evidence of stable remanent magnetization in response to AF demagnetization.

The magnetization of cores from site 9 shifted away from the earth's magnetic field direction during step-cleaning and away from the mean direction of the group. They exhibited a good end-point and are best grouped after demagnetization in a peak field of 250 Oe (Table 1, Fig. 3). As this end-point direction is distinct from that of the group mean, the statistics for

TABLE 1. Site and group directions before and after AF cleaning, group pole positions after AF cleaning

			NRM				Cleaned					
Site	*N*	*n*	*D*	*I*	*K*	δ°	AF	*D*	*I*	*K*	δ°	δ°/R
1	7	7	289	80	733	3.0	250	274	67	615	3.3	1.2
2*	10	10	228	78	67	9.9	250	224	60	991	2.6	.8
3	7	7	247	79	143	6.8	200	242	77	76	9.3	3.5
4	6	3	69	55	29	15	500	138	72	89	8.6	5.0
5	6	6	130	82	21	18	250	193	72	199	5.7	2.3
6	6	4	190	63	1.6	64	350	210	74	138	6.9	3.4
7	10	9	142	21	3.3	45	350	193	75	169	6.2	2.1
8	8	8	85	82	24	16	350	209	70	232	5.3	1.9
10	7	6	186	56	1.8	61	250	212	74	122	7.3	3.0
13	6	3	290	84	28	15	400	209	79	20	18	10
All cores		63	164	79	4.7	38		218	73	41	13	1.6
All sites		10	160	78	9.0	27		213	75	48	12	3.7
Mean (North) Pole Position			114°W	28°N	$K = 17$	$\alpha_{95} = 12$						
9	6	6	133	59	2.3	54	250	143	42	39	13	5.3
Virtual (North) Pole Position			64°W	5°S								

*Magnetic direction reversed for use in statistics.
NOTES: N = number of cores collected at site; n = number left after excluding those that are inhomogeneously magnetized ($\theta > 25°$). D = declination, I = inclination, $\delta°$ = angular standard deviation, K = Fisher's estimate of precision, AF = peak alternating cleaning field in oersteds, δ°/R = standard error of the mean, α_{95} = half angle of cone of confidence.

this site are listed separately in Table 1. The chemistry of the dike at site 9 is identical to that of the remaining dikes.

Site 2 is reversely magnetized relative to the group. This provides supporting evidence for the isolation of a stable primary TRM in the Molson dikes.

Silicate Petrography

The mineralogy of the dikes has been classified as magmatic, late-stage magmatic, and deuteric hydrothermal. The 'deuteric hydrothermal' alteration may be due to low-grade regional metamorphism. Olivine, orthopyroxene, clinopyroxene (augite), bytownite–andesine (with 0.2 to 4.0 mol % orthoclase), and hornblende, represent initial phases in both coarse- or fine-grained rocks, but form phenocrysts in aphanitic rocks (Fig. 4). Olivine, as an accessory phase, is generally recognized as serpentine and magnetite pseudomorphs (of corroded olivines) in most rocks; at site 7 olivine occurs unaltered except for resorbed grain borders which show alteration to serpentine and magnetite. The primary phases show varying degrees of crystal fractionation features, such as oscillatory and discontinuous zoning (plagioclases), and resorption and crystal mantling (olivine rimmed by pyroxene, orthopyroxene by clinopyroxene).

In the late-stage magmatic assemblages the amount of hornblende increases with the appearance of biotite; both minerals are associated with magnetite, and interstitial albite (rarely microcline) and quartz intergrowths (symplectite or granophyre) are present. In extreme cases albite rims plagioclases that have bytownite cores (An_{83}), and quartz forms discrete, commonly rounded and resorbed grains. At this stage quartz is commonly intergrown with opaque minerals and biotite. Most, but not all coarse-grained dikes show a central granophyric zone; the granophyre may occur throughout the dike. The degree and amount of granophyric development seems to be unrelated to the width of the dikes. At site 4 where quartz-albite symplectite is present throughout the dike, a 1-in. (2.5 cm) wide quartz porphyry dikelet of similar composition was found 20 ft (6.1 m) from the main dike. Indeed, aphanitic, thin, satellite dikes faithfully reflect the composition of the main dikes (*e.g.*, Table 2, site 10).

All 'post-magmatic' phase changes in the present rock appear to indicate autochthonous hydrothermal alteration (Fig. 5) rather than recrystallization due to regional greenschist metamorphism. The degree of hydrothermal alteration seems to be related directly to the degree of differentiation as indicated by plagioclase compositions. Rocks with plagioclase ranging from bytownite (An_{80}) to albite (An_{10})

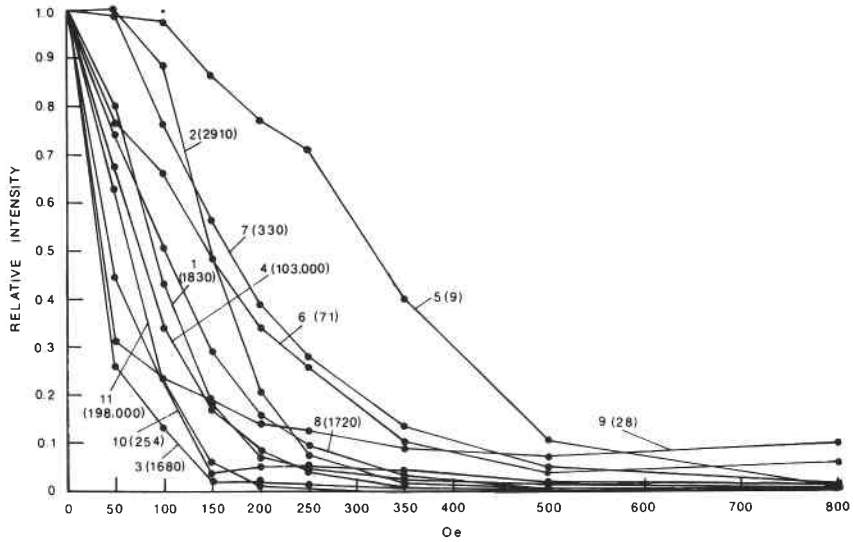


FIG. 2. Normalized AF demagnetization curves of one specimen from each site. Unbracketed number identifies site from which specimen was derived; bracketed number indicates initial intensity in e.m.u. $\times 10^{-6}$.

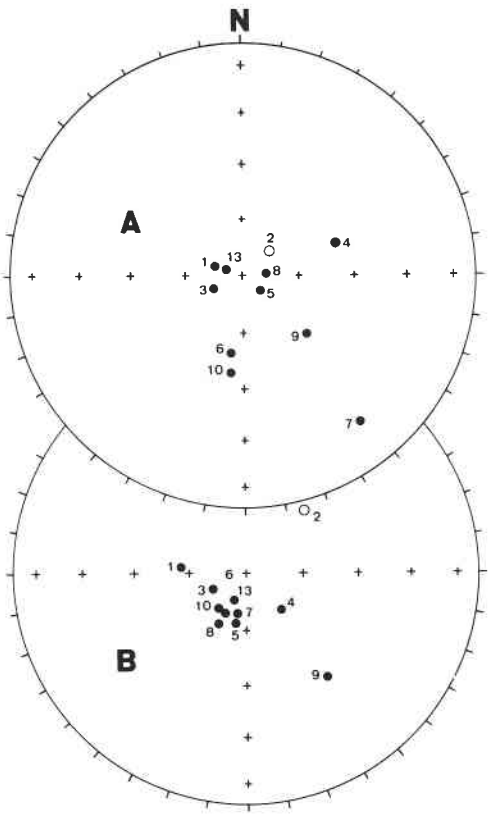


FIG. 3. Equal area projection of site mean magnetization directions before (A), and after (B), optimum AF demagnetization. Solid circles: north seeking directions down; open circle: north seeking direction up.

and which develop trace amounts of microcline ($An_{10}Ab_{45}Or_{45}$, or $An_5Ab_{70}Or_{25}$) show the greatest degree of deuteric (hydrothermal) alteration. Alteration is minimal in diabases in which the plagioclase zoning is more limited (An_{70} to An_{50}). Mineral phases formed or accentuated during the 70-50 hydrothermal state are epidote, sericite, chlorite, carbonate, sphene, fibrous amphibole (uralite), serpentine, and talc. All high-temperature phases are altered to a degree, but alteration is nearly 80% in assemblages in which late-stage magmatic effects are pronounced. Hydrothermal alteration appears to be pervasive but variable in coarse-grained dikes, whereas similar alteration products in aphanitic dikes are confined to hair-line veins.

The dike at site 7, in addition to being a multiple injection, is also reversely zoned in that coarse olivine and orthopyroxene are concentrated in the center of the dike (15% MgO). This dike may demonstrate flow differentiation as a result of which the first phases to crytallize collect in the center as a crystal mush, leaving the liquid residue toward the walls (Bhattacharji and Smith 1964). NRM directions for this site were highly dispersed, but this dispersion decreased greatly with AF cleaning. A single core from this site, belonging to the 'second phase' of injection, though stably magnetized, differed in direction by more than three standard deviations from the mean site direction and so

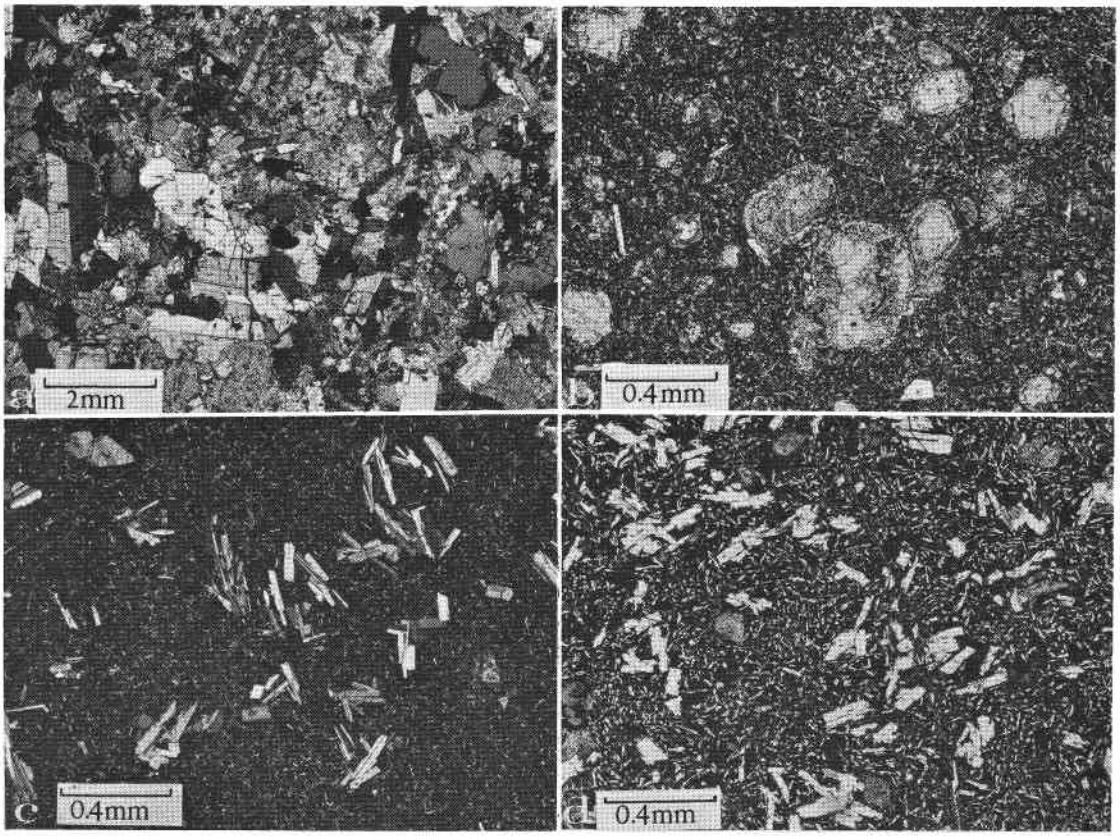


FIG. 4. Textural variation of chilled and gabbroic dikes.
(a) gabbroic texture at site 11; corroded bytownite, bastite (orthopyroxene), completely altered olivine, clinopyroxene, and basaltic hornblende. Crossed nicols. (202461-C)[1]
(b) aphanitic 1.5-ft (0.46-m) wide satellite dike at site 11; orthopyroxene with olivine cores rimmed by clinopyroxene in a matrix of plagioclase fibrolites and clinopyroxene. Plane light. (202461)
(c) aphanitic, 10-in. (25-cm) wide dike at site 2; serrated andesine laths, corroded orthopyroxene, and clinopyroxene in a matrix of the same minerals. Crossed nicols. (202311-A)
(d) coarser-grained center of 10-in. (25-cm) wide diabase dike at site 2; irregular, corroded andesine laths and prismatic clinopyroxene rimmed by green hornblende in a matrix of the same minerals; vein comprises fibrous minerals, apatite, and feldspar. Plane light. (202461-N)

was not included in the site mean. Interestingly, the plagioclase of this sample appears to be noticeably stressed, the twinning is rather poor, and grains exhibit wavy extinction and are broken and rotated.

Opaque Petrography

The opaque mineral content of the dikes ranges from 3% to trace amounts. Rocks with abundant magnetite and magnetite with ilmenite lamellae often include up to ½% pyrite, pyrrhotite, and in a few places, chalcopyrite. Stages and textures of high-temperature exsolution of ilmenite, and oxidation of magnetite (hematite) vary from specimen to specimen. In some rocks magnetite produced from hydrothermal alteration of olivine or orthopyroxene, is the dominant iron oxide phase. In granophyric rocks magnetite is commonly associated with quartz, biotite, albite (in places microcline), and rarely, hornblende. Additionally, sphene and fibrous ferrous silicates in rocks exhibiting a high degree of alteration are pseudomorphic after magnetite–ilmenite euhedra (Fig. 6); this implies a

[1]Numbers in parentheses are catalogued numbers in the photographic library of the Geological Survey.

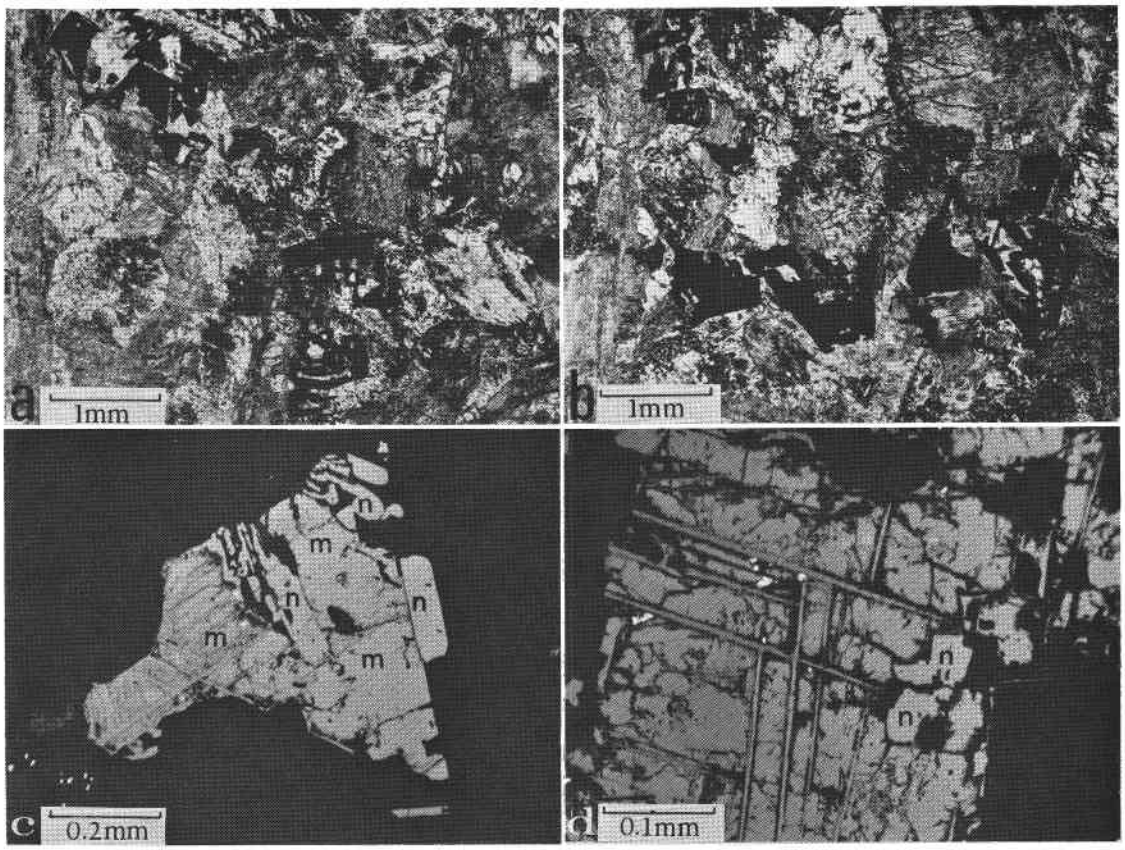


FIG. 5. Opaque mineralogy at site 4.
(a) skeletal crystal forms of magnetite–ilmenite in a matrix of epidotized-sericitized zoned bytownite clinopyroxene (altered to hornblende + magnetite–hematite dust), basaltic and fibrous hornblende, and corroded interstitial quartz. Crossed nicols. (202461-I).
(b) like Fig. 5a: quartz as inclusions in skeletal magnetite and ilmenite framework. Crossed nicols. (202461-K).
(c) titaniferous magnetite (m) with ilmenite lamellae and clear grains of ilmenite (n); interlamellar areas (m) contain hematite alteration near ilmenite lamellae and altered titaniferous magnetite away from ilmenite lamellae. Plane light.
(d) ilmenite lamellae in altered 'titaniferous magnetite'; portions labelled n are ilmenite; 'titaniferous magnetite' areas are mostly ilmenite or ilmenite-hematite. Plane light.

movement of Fe and Ti away from the primary oxide phase.

Stronger NRM is readily related to high contents of opaque minerals regardless of the degree of alteration in the rock. The degree of hydrothermal alteration, however, was found to vary a great deal among cores of a site, and it is this variation that may produce within-site dispersion of NRM. Aphanitic dikes with a low magnetite content in which magmatic differentiation is slight and in which hydrothermal alteration is confined to veins, generally exhibit moderate magnetic intensities and good groupings. Presumably the very small grains have only a minor viscous component.

Chemistry and Classification

The analysis of Table 2 were performed by staff of the analytical chemistry section of the Geological Survey of Canada. Spectrochemical methods QN-12B and QN-12DR were used for Sr, Ba, Cr, Zn, Ni, Cu, Co, La, and V, and are expected to be accurate to within $\pm 15\%$ of the value reported. Zn, Pb, and Ag (method QN-13) are accurate to within $\pm 30\%$ of the value reported. Major elements were determined by

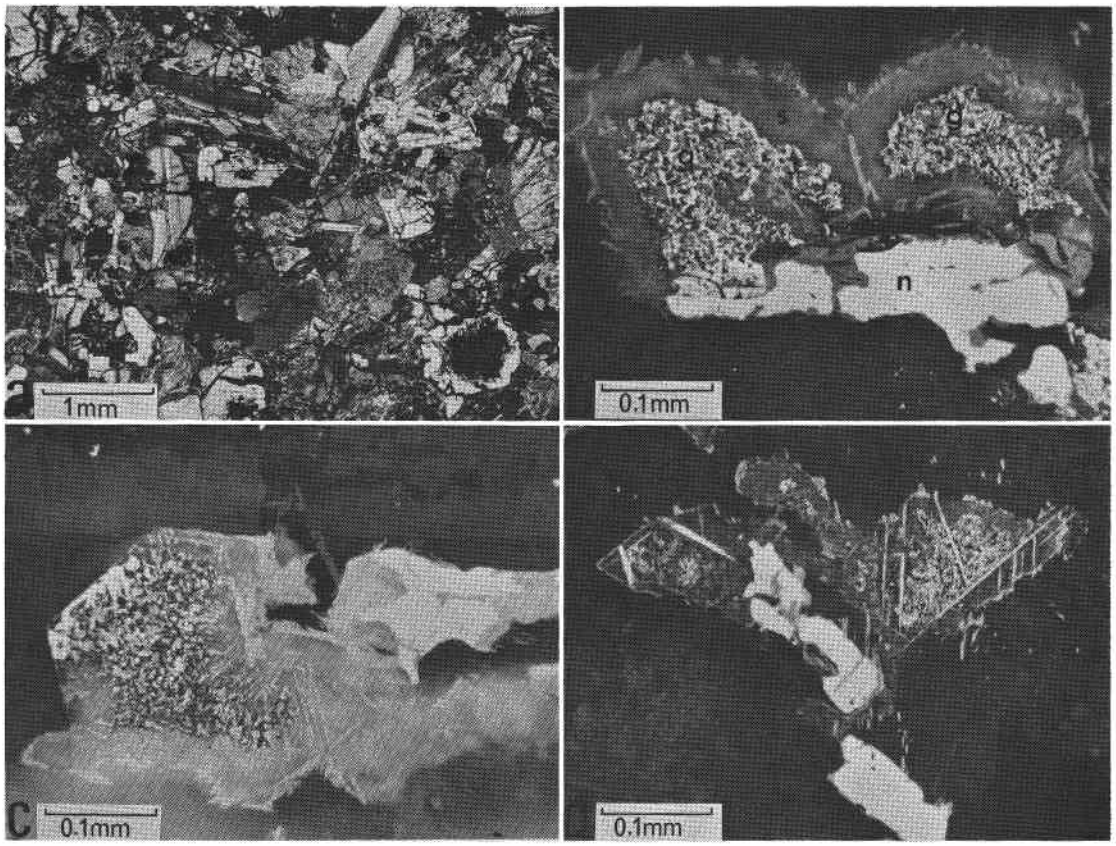


FIG. 6. Opaque mineralogy at site 10.
(a) zoned, bytownite–albite reactive with chlorite and fibrous hornblende along grain edges and fractures; relatively unaltered ortho- and clinopyroxene and pale brown idiomorphic hornblende. Crossed nicols. (202461-D)
(b) ilmenite (n) and sphene (s) enclosing a granular matrix of ilmenite + magnetite + silicate (g). Plane light.
(c) as in Fig. 6b, but ilmenite lamellae are visible in alteration product (sphene and other silicates), which appears to have a pseudomorphic magnetite crystal habit. Plane light.
(d) as in Fig. 6c; granular magnetite + ilmenite + silicate has fabric parallel to one of the ilmenite lamellae directions (short diagonal). The granular fabric is interpreted as exsolution of ilmenite-rich magnetite, during a secondary oxidation phase; exsolution of ilmenite lamellae is a primary high-temperature oxidation feature.

rapid methods (XRF not fused), and conventional chemical, differential colorimetric (SiO_2 and Al_2O_3), and atomic absorption (MnO, CaO, K_2O, MgO, Na_2O, Rb). The results of 30 of these analyses are grouped and averaged in Table 2. Normative calculations are plotted (Figs. 7 and 8) according to the scheme and classification of Irvine and Baragar (1971).

The rocks are tholeiitic basalts (Figs. 7 and 8). Granophyric portions of dikes represent extreme magma differentiation—for example, sites 12 and 13, whose coarse granophyric centers approach the calc-alkaline dacite field of composition from an initial tholeiitic basalt. A less pronounced but systematic trend toward the calc-alkaline andesite field of composition is not shown by rocks of site 4 (4A to 4G, Fig. 8). It is interesting that the bulk normative anorthite content increases slightly (*ca.* An_{30} to An_{35}) in the granophyre at sites 12 and 13 (Fig. 7). At site 4, however, granophyre 4c (Fig. 8) does show a decrease of normative anorthite in the plagioclase from *ca.* An_{30} to An_{20}.

The major elements at site 4 show a normal

TABLE 2. Major elements (wt. %) and trace elements (ppm) of Molson diabase dikes

Site Number (Fig. 1)	1	2(3)	3	4A	4B	4C	4D	4E	4F	4G	5&6	7	8	9(3)	10A	10B(2)	11A	11B(4)	12A	12B	13A	13B
Dike width (in feet)	4.5	1	3				60				4	30	1.2	90	2	200	1.5	200	>200		>200	
SiO_2	49.8	48.8	49.4	51.7	48.4	49.5	51.1	49.7	51.0	50.5	46.8	47.1	49.8	49.3	48.3	49.2	48.7	51.7	48.8	49.1	49.5	48.8
Al_2O_3	14.0	12.8	12.4	12.4	13.2	11.3	13.5	13.2	9.5	13.6	12.0	10.6	14.3	12.3	14.8	13.4	12.1	14.5	14.4	16.9	13.7	21.1
Fe_2O_3	2.6	2.9	3.2	1.5	2.0	5.4	1.7	1.6	1.6	1.2	2.6	1.7	2.6	3.3	2.1	2.2	3.1	1.7	1.5	3.9	0.05	0.05
FeO	8.5	12.5	7.8	6.6	7.3	10.1	8.0	7.5	6.9	7.4	8.2	9.0	8.6	8.3	7.8	7.9	7.5	7.9	9.7	9.0	10.4	9.0
MgO	8.3	5.5	9.3	11.5	9.1	6.2	9.4	10.4	11.0	10.5	12.3	15.8	7.6	8.2	9.3	10.5	11.7	10.2	6.7	1.6	10.2	1.6
CaO	11.9	9.9	12.1	14.5	11.6	9.1	10.8	11.4	13.9	12.4	10.5	9.0	11.3	10.6	11.0	11.3	10.8	11.3	12.0	9.1	11.3	8.3
Na_2O	1.4	2.1	1.3	1.1	1.3	2.4	1.4	1.1	1.0	1.1	1.1	0.9	1.7	1.9	1.0	1.1	1.4	1.2	1.6	3.1	1.6	5.3
K_2O	0.2	0.8	0.4	0.3	0.5	0.6	0.2	0.5	0.5	0.5	0.4	0.8	0.3	0.5	0.2	0.4	0.2	0.3	0.4	0.6	0.5	0.4
TiO_2	0.97	1.7	0.92	0.41	0.48	1.1	0.65	0.46	0.51	0.44	0.74	0.69	0.99	0.79	0.81	0.81	0.83	0.73	1.1	1.7	0.87	1.4
P_2O_5	0.07	0.17	0.06	0.02	0.03	0.11	0.05	0.07	0.04	0.33	0.05	0.05	0.07	0.07	0.07	0.06	0.06	0.06	0.08	0.14	0.07	0.16
MnO	0.19	0.24	0.19	0.08	0.17	0.23	0.20	0.17	0.18	0.17	0.21	0.19	0.20	0.20	0.19	0.18	0.20	0.19	0.19	0.18	0.16	0.13
S	0.10	0.07	0.08	ND	ND	ND	0.06	ND	ND	ND	0.05	0.06	0.08	ND	ND	ND	0.09	ND	0.10	0.13	0.09	0.00
CO_2	0.06	0.05	0.04	0.00	0.00	0.52	0.11	0.00	0.00	0.00	0.85	0.00	0.36	0.00	0.00	0.03	0.05	0.04	0.00	0.00	0.00	0.00
$H_2O(T)$	1.8	1.6	2.6	1.7	2.3	2.8	2.5	2.2	1.7	2.3	3.9	1.7	2.5	2.5	2.9	3.2	3.5	2.6	1.7	2.1	2.5	2.5
Zn	100	100	150	50	60	90	170	60	50	50	80	500	210	60	70	60	100	70	60	100	60	40
Pb	<.5	3	2	<.5	1	3	2	1	1	<.5	1	4	3	2	1	1	3	1	.6	1	4	.7
Cu	150	260	150	50	60	180	100	70	60	50	140	100	180	140	130	120	130	110	ND	ND	ND	ND
Ni	120	90	140	150	140	70	140	160	150	150	260	560	120	120	260	240	250	260	ND	ND	ND	ND
Cr	340	90	490	1800	420	10	510	650	870	1900	740	1540	220	110	700	530	730	810	ND	ND	ND	ND
Sr	80	120	130	60	100	120	70	90	70	70	60	40	70	120	80	80	170	90	ND	ND	ND	ND
Ba	60	150	80	40	90	120	50	50	60	50	60	60	40	60	40	40	50	30	ND	ND	ND	ND
Zr	130	180	130	<30	60	170	120	60	50	40	ND	110	130	80	90	70	120	80	ND	ND	ND	ND

Rb ranges from undetectable quantities to 20 ppm. Ag is present generally in amounts less than 50 ppb, but 4 dikes have a range from 60 to 200 ppb. Co in 8 samples ranges from 40 to 80 ppm ($\bar{x}$ = 60 ppm); La in 6 samples ranges from 280 to 380 ppm (x = 350 ppm); V in 8 samples ranges from 250 to 420 ppm ($\bar{x}$ = 310 ppm). Dikes less than 5 ft wide are aphanitic; sites 4, 7, 12, and 13 are coarse grained and differentiated: sites 9, 10,and 11 are coarse grained and massive. Site 2 is reversely magnetized. Number in brackets beside site indicates number of samples analyzed and averaged for that column. Samples 4A to 4G represent a series from margin to margin (see fig. 4). 10A and 11A are satellites to 10B and 11B respectively. 12A and 13A are chilled margins, 12B and 13B are granophyric centers.

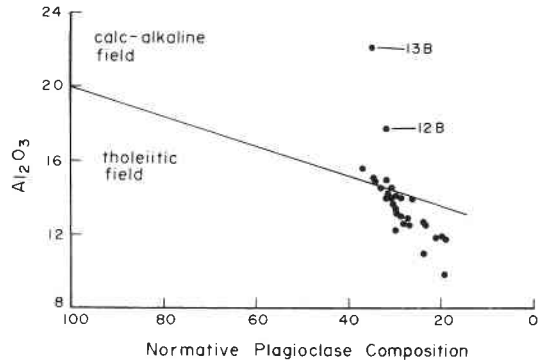


FIG. 7. Plot of wt.% Al_2O_3 versus normative plagioclase of diabase dikes. The line divides the tholeiitic from the calc-alkaline field as proposed by Irvine and Baragar (1971, p. 536). Specimen 12B and 13B are coarse-grained granophyric centers; their chilled margin 'pairs', however, fall well within the tholeiitic field. Remaining dikes that fall in calc-alkaline field are less than 3 ft (1 m) wide.

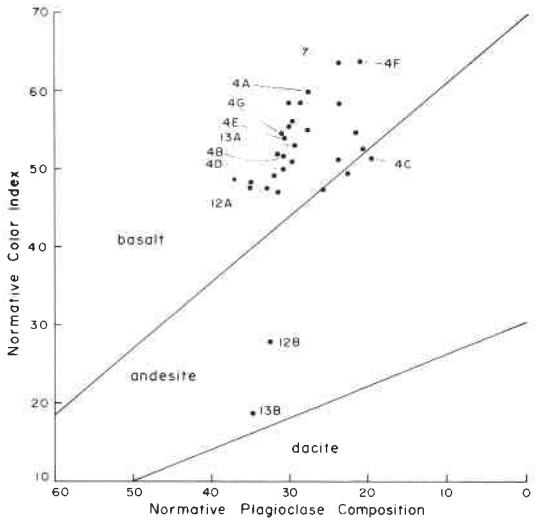


FIG. 8. Plot of normative color index versus normative plagioclase of diabase dikes. The lines divide the basaltic, andesitic, and dacitic fields of composition as proposed by Irvine and Baragar (1971, p. 538). Numbers refer to sample sites (Fig. 1 and Table 2). Site 12 and 13 are strongly differentiated with a constant normative plagioclase composition. Site 4 is less strongly differentiated (4A to 4G), but shows a decrease in normative plagioclase composition.

sequence of differentiation toward an acidic fraction (4c, Table 2). Especially noteworthy at site 4 is the increase in TiO_2, total iron, and oxidation state of iron with increasing granophyre formation. Trace elements of this site show a pronounced trend: Cr and Ni decrease, and Zn, Cu, Zr, Ba, Sr increase with increasing acidic differentiation (Fig. 9). Site 7 carries 15.8% MgO and anomalously high Ni (560 ppm), Zn (500 ppm), and Cr (1540 ppm) relative to the other dikes. Judging from the low sulfur content of the rock at site 7 (600 ppm), most of the Ni and Zn must occur in the silicates.

Site 2 is a splay of 1-ft (0.3 m) thick dikes occupying a zone at least 100 ft (31 m) wide. The dikelets are of uniform composition, reversely magnetized, and are the most acidic of all of the dikes (except for granophyre) at sites 4, 12, and 13. The rock at site 2 is also the most TiO_2-rich (1.7%) of all the intrusions (equal to granophyre 12B). It is possible that the magnetization of this site is the result of self-reversal following the mechanism suggested by Ozima and Larsen (1970).

Age Determinations

A number of K/Ar age determinations on biotite and hornblende in the area of the dikes indicate that Archean metamorphism and plutonic activity ceased 2.45 to 2.65 b.y. ago. (See Table 3, which contains unpublished data from the Geochronology Section of the Geological Survey of Canada, and from Lowdon (1961, p. 51), Lowden *et al.* (1968, p. 72), and Wanless *et al.* (1965, p. 77).) However in the area between the Churchill–Superior tectonic boundary, and a line that begins at the outlet of Lake Winnipeg and trends northeastward parallel to the Churchill–Superior boundary, biotites in most rocks record a thermal event 2.2 to 1.8 b.y. ago (isotopic age map of Canada (Geological Survey of Canada map 1256A) and Table 3). The effects of this thermal event extend 75 miles (120 km) southward into the Superior Province and into the area of the present dikes (except site 1) and appear to be a metamorphic front, emanating from the Churchill Province. Hornblendes in this area, however, continue to record Archean time (*e.g.*, amphibolite dikes, Table 3).

With the exception of site 7 (Table 3) the whole-rock K/Ar age of the dikes ranges from 1.3 to 1.7 b.y. The range in K/Ar age probably reflects variation in the loss of argon. All of the dikes, including granophyric portions, have rather low potash contents for continental tholeiites. Site 7, which is a composite olivine-bearing dike, contains 0.08% K and yielded an age of 2.4 b.y. on a uralitized sample. A very small quantity of excess argon in this sample

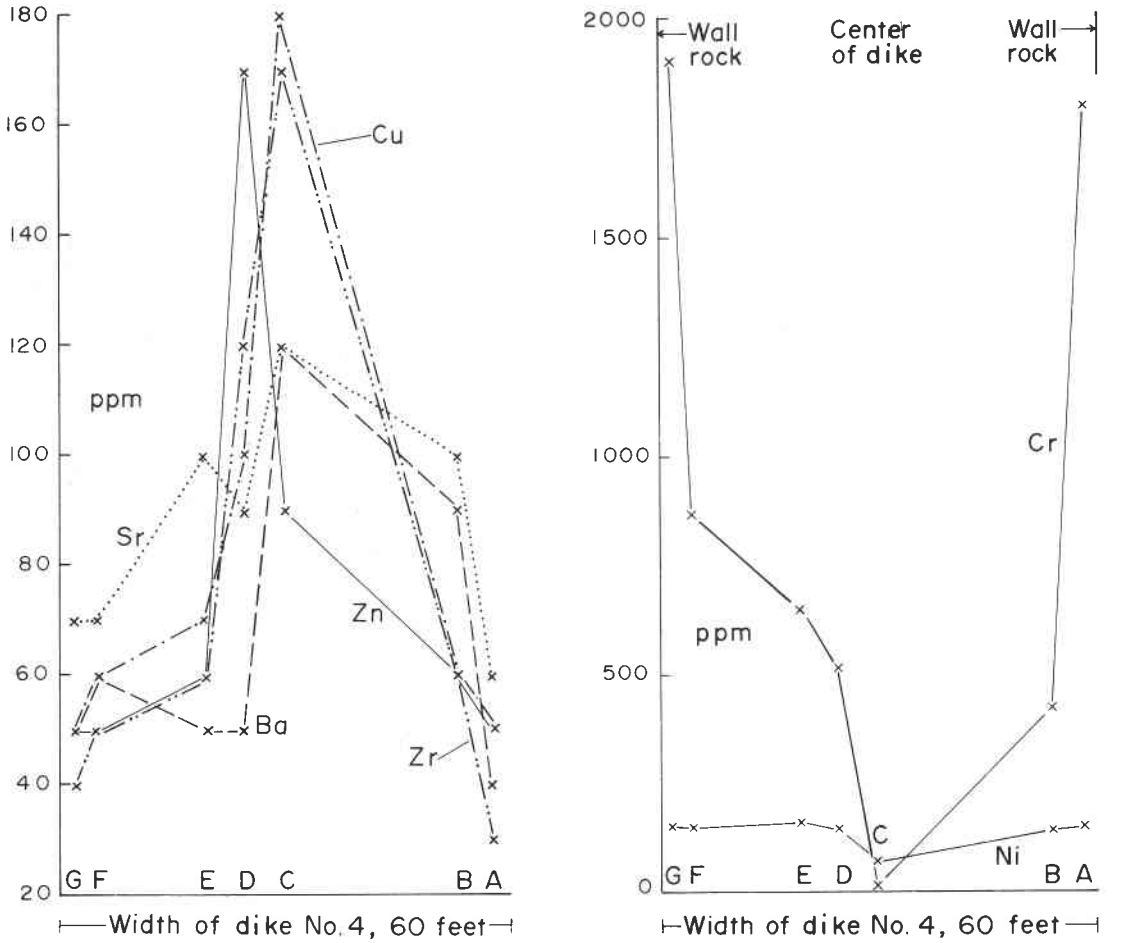


FIG. 9. Plot of trace elements (ppm) versus position of samples within dike at site 4. Analyses appear in Table 1 (4A to 4G).

could account for the apparently anomalously great age. An age of nearly 3.0 b.y. has been obtained from material of site 2 (R. K. Wanless, personal communication, 1974), again suggesting the presence of excess argon.

Discussion

The presence of one reversely magnetized dike in the midst of nine normally magnetized dikes may suggest that the magnetization of the Molson dikes was acquired at the time of their intrusion. The dikes do contain evidence of widespread alteration, but it is unlikely that a field reversal during a period of low-grade regional metamorphism would produce sharply defined normally and reversely magnetized dikes. If, as suggested previously, the reversed site is the result of a 'self reversal' mechanism, this would probably be operative during a late magmatic stage. Furthermore, some of the dike material is extremely fresh, and this material has the same magnetization direction as the altered dike material. The alteration products in these dikes may be simply a deuteric effect. Whether or not the magnetization is primary (magmatic) or related to a significantly later low-grade regional metamorphic event, the K/Ar ages may provide a clue to the age of magnetization. It has been found that paleomagnetism systematically 'looks through' the whole-rock ages of basic igneous rocks (Fahrig and Jones 1969). If this is true in the present case, and if we discount those K/Ar ages thought to reflect the presence

TABLE 3. Potassium–argon whole-rock ages*

Location	% K	% Radiogenic Ar	Notes	Age (m.y.)
1	0.168	89.0	Whole-rock, diabase, aphanitic; secondary amphibole	1646 ± 111
7	0.087 0.088	92.1	Whole-rock, diabase, micro-granular margin; uralite	2393 ± 270 2368 ± 270
10	0.178	94.2	Whole-rock, aphanitic satellite dike of coarse-grained parent	1337 ± 96
12A	0.38	81.0	Whole-rock, porphyritic chilled margin to a granophyric diabase	1445 ± 180
13A	0.45	72.0	Whole-rock, microgranular margin to a coarse granophyric dike	1280 ± 180
Cross Lake[1]	8.25	100.	Biotite from quartz monzonite	2100 ± 70
Cross Lake[2]	7.28	97.0	Biotite from garnetiferous paragneiss	2135 ± 125
Cross Lake[3]	7.28	96.0	Biotite from garnetiferous paragneiss	2065 ± 125
Playgreen Lake; near site 8	7.80	92.0	Biotite from quartz monzonite	2190 ± 125
Six miles south of Norway House	7.70	99.6	Biotite from amphibolite dike	2065 ± 48
on the Gunisao River	0.83	98.8	Hornblende from same dike	2684 ± 63

*Age determinations carried out by the Geochronology Section of the Geological Survey of Canada; [1]Wanless *et al.*(1965). [2]Lowdon *et al.* (1968). [3]Lowdon(1961). Unreferenced ages, Wanless *et al.* (in preparation).

of excess argon, the K/Ar ages suggest an age of magnetization in the range 1800–2000 m.y.

The Molson pole at 114°W, 28°N is close to a large group of North American Precambrian poles whose age is thought to be late Aphebian. These are the Sudbury Norite (Sopher 1963), Otish Gabbro (Fahrig and Chown 1973), Spanish River alkaline complex (Roberston and Watkinson 1974) Kaminak diabase (metamorphosed)[2] and Kaminak lamprophyre.[2] The paleomagnetic poles for all of these units lie in the southern United States, and together they form an anchor, or possibly a cross-over in a loop, for the paleomagnetic wandering curve of the Superior Orogen during Aphebian time.

The virtual pole obtained from site 9 at 064°W, 5°S lies a considerable distance southeast of the mean Molson pole. Its position relative to the Precambrian polar wandering curve suggests that its magnetization is older than that of the Molson. On the other hand, the magnetization direction of site 9 lies in a plane that includes both the normal and the reversed Molson directions, so that its direction may reflect the presence of both normal and reversed components.

[2]Christie, K. W., Davidson, A., and Fahrig, W. F. The paleomagnetism of the Kamiak diabase dikes: no evidence of Hudsonian tectonics. (Manuscript in preparation).

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